Effects Validations

August 2, 2025

This document validates the calculated effects by demonstrating their reasonableness through comparison with import vectors. Effects should closely resemble import vectors, differing only where consumer price effects are not captured by imports alone.

I focus primarily on direct effects, which incorporate only imports and consumer expenditures and should therefore closely align with import vectors. Indirect effects, which include complex domestic production processes, are addressed tangentially. The analysis proceeds in two stages: first demonstrating overall similarity between direct effect and import vectors, then explaining discrepancies through the presence of large non-consumer imports (oil, gas, 'Other').

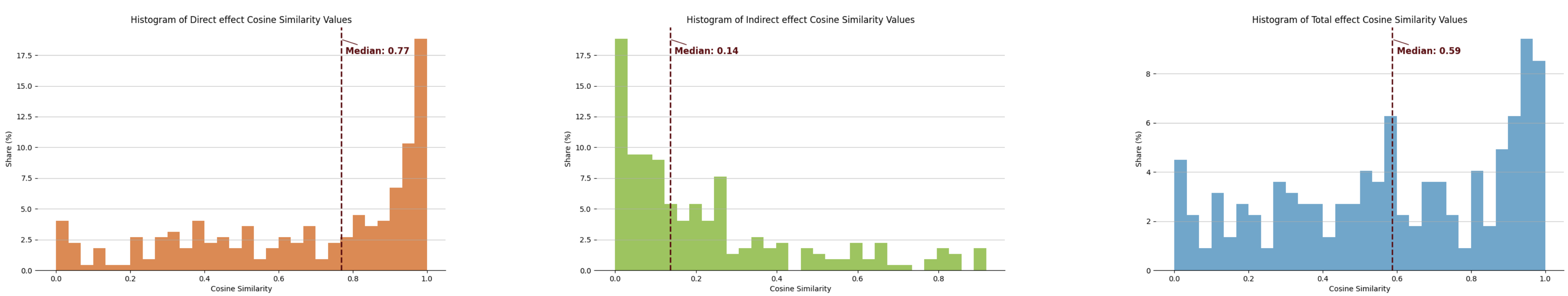
Section 1 looks at how similar the direct effects vector for a country is to its import vector. Section 2 tries to cluster countries together by the kMeans similarity of their import vs their effect vectors. Section 3 provides a short case study of two earlier identified 'probleatic' countries, the UK and Israel, to look at exactly what might be driving the differences between their import and effect vectors.

1 Cosine Similarity Analysis

I compute cosine similarities between import and effect vectors within countries, expecting direct effects to closely resemble import vectors. Figures **??**, **??**, and **??** confirm this expectation, showing direct effects achieve high cosine similarity with a median around 0.77.

Each histogram represents a country grouping. Dispersion in within-country cosine similarities for direct effects stems from unique import structures featuring either high-impact or low-impact products. I provide country-level narratives for the bottom 4 bins explaining low cosine similarities.

Notably, indirect effects exhibit much lower cosine similarity, which transfers to total effects, because they incorporate domestic production processes absent from import vectors.



(a) Direct Effects (b) Indirect Effects (c) Total Effects

Figure 1: Within Country Cosine Similarity Distributions

1.1 Country-Level Analysis

This section examines import structures of countries with the lowest direct effect-import cosine similarities to understand whether discrepancies arise from importing goods with high consumer spending shares (e.g., clothing) or goods with minimal direct effect impact.

Table **??** presents the bottom 4 bins of direct effect cosine similarities. These within-country similarities compare each commodity's import share of total country imports against its direct effect share of total country direct effects.

Table 1: Bottom 4 Cosine Similarity Bins for Direct Effect

Country	ISO	Cosine Similarity	Bin No.
Libya	LBY	0.00022	0
Equatorial Guinea	GNQ	0.00086	0
Sint Maarten	SXM	0.00216	0
Guyana	GUY	0.00523	0
Djibouti	DJI	0.01732	0
Tokelau	TKL	0.01951	0
Venezuela	VEN	0.02336	0
British Virgin Islands	VGB	0.02351	0
DR Congo	COD	0.03292	0
Bermuda	BMU	0.05227	1
Chad	TCD	0.05314	1
Antigua and Barbuda	ATG	0.05339	1
Cook Islands	COK	0.06149	1
Angola	AGO	0.06392	1
Tajikistan	TJK	0.09388	2
Nigeria	NGA	0.10846	3
Azerbaijan	AZE	0.11493	3
Aruba	ABW	0.12334	3
Guadeloupe	GLP	0.12585	3
Kazakhstan	KAZ	0.15629	4

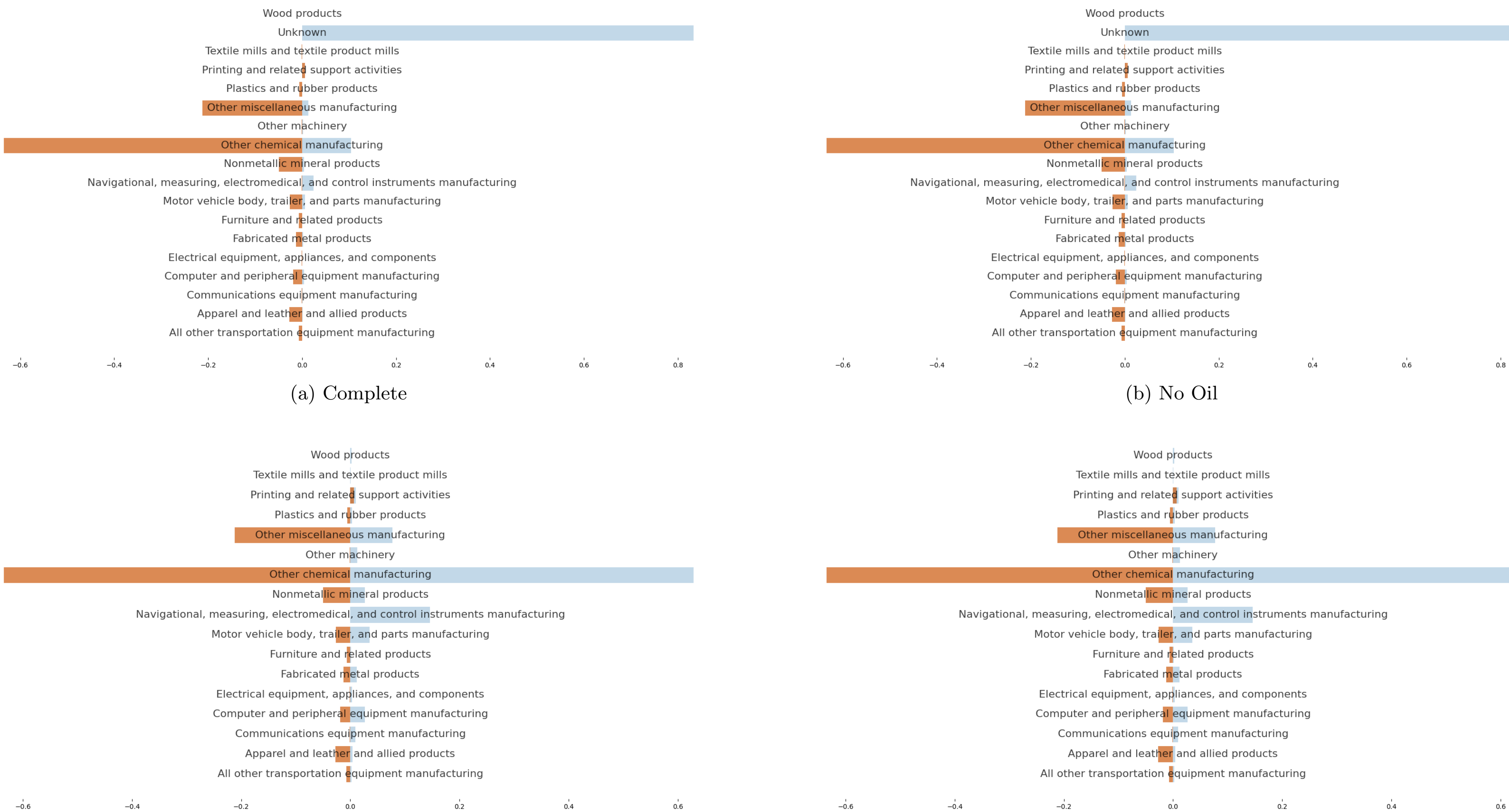
The following figures compare import structures with direct effects for bottom-bin countries using four panels: (a) all non-zero import commodities, (b) excluding oil, (c) excluding 'Other' category, and (d) excluding both oil and 'Other'. This progressive exclusion tests whether removing non-consumer goods improves alignment quality.

Results confirm that non-consumer goods drive low cosine similarities. After excluding oil and 'Other', remaining mismatches primarily involve commodities like nonferrous metal production.

Analysis reveals that low cosine similarity often stems from countries importing large quantities of single goods (oil, nonferrous metals, Unknown/Other). Since cosine similarity uses shares, dominant single goods reduce similarity scores. Even after removing Oil and Other categories, mismatches persist between import structure and direct effects.

Table **??** presents the top 6 commodities with largest import-direct effect share differences. The left panel shows commodities with higher import than direct effect shares; the right panel shows the reverse. Differences are calculated as import share minus direct effect share, with the 'Diff.' column showing average differences across the bottom 4 cosine similarity bins. These differences clearly reflect relative personal consumption expenditures between commodities.

Figure 2: Aruba Imports vs. Direct Effects Shares



(a) Complete

(b) No Oil

(c) No Other

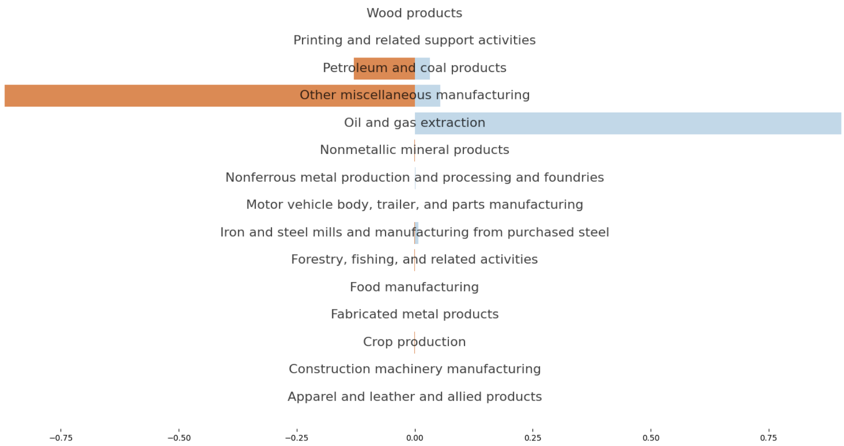
(d) No Oil, No Other

Table 2: High (Low) Import Shares vs. Low (High) Direct-Effect Shares

High Import / Low Direct Effect				Low Import / High Direct Effect			
Rank	**Commodity Description**	**Diff.**	**PCE Share (pctile)**	**Rank**	**Commodity Description**	**Diff.**	**PCE Share (pctile)**
1	Nonferrous metal production and processing and foundries	0.312	0.000008 (32.8%)	1	Beverage manufacturing	0.154	0.007 (80.0%)
2	Aerospace product and parts manufacturing	0.108	0.00007 (39.3%)	2	Food manufacturing	0.144	0.0313 (96.4%)
3	Basic chemical manufacturing	0.084	0.000078 (38.6%)	3	Other miscellaneous manufacturing	0.102	0.005 (73.57%)
4	Iron and steel mills and manufacturing from purchased steel	0.0669	0.000078 (37.8%)	4	Crop production	0.093	0.0054 (72.9%)
5	Mining, except oil and gas	0.0582	0.0000098 (33.6%)	5	Apparel and leather and allied products	0.474	0.009 (81.4%)
6	Electrical equipment, appliances, and components manufacturing	0.0370	0.00359 (66.4%)	6	Fabricated metal products	0.366	0.0016 (59.3%)

Notes: Goods ranked by the average difference in import share and direct effect share for the bottom 4 bins of cosine similarity between the direct effect vector and the import vector.

Figure 3: Angola Imports vs. Direct Effects Shares



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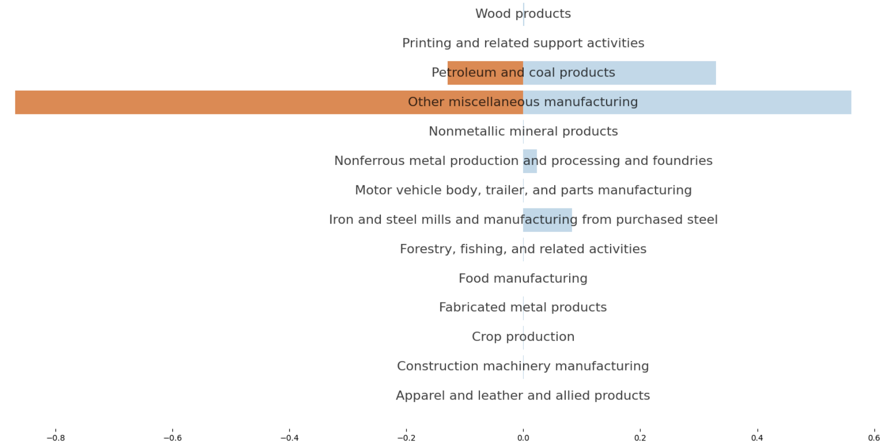
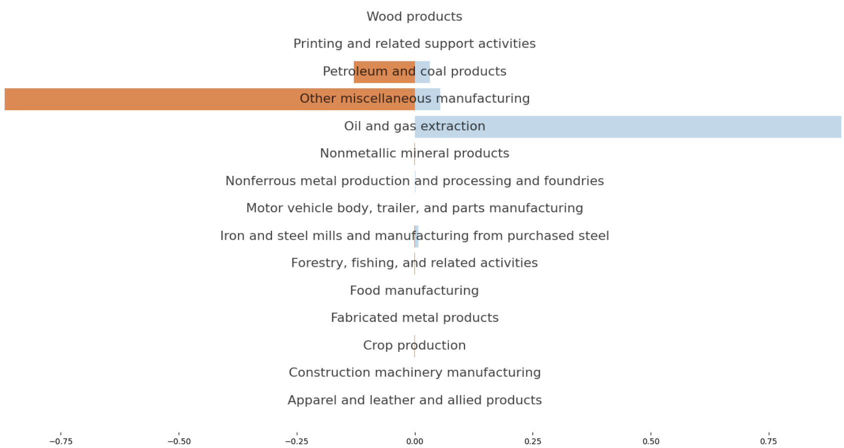
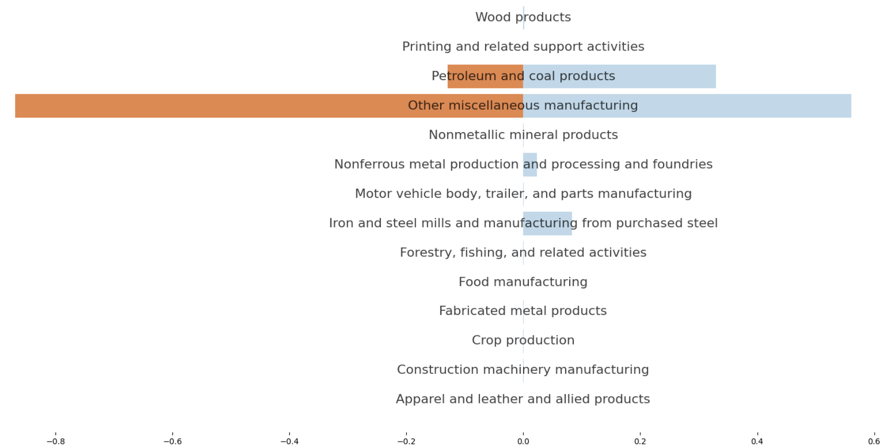


(b) No Oil



(c) No Other

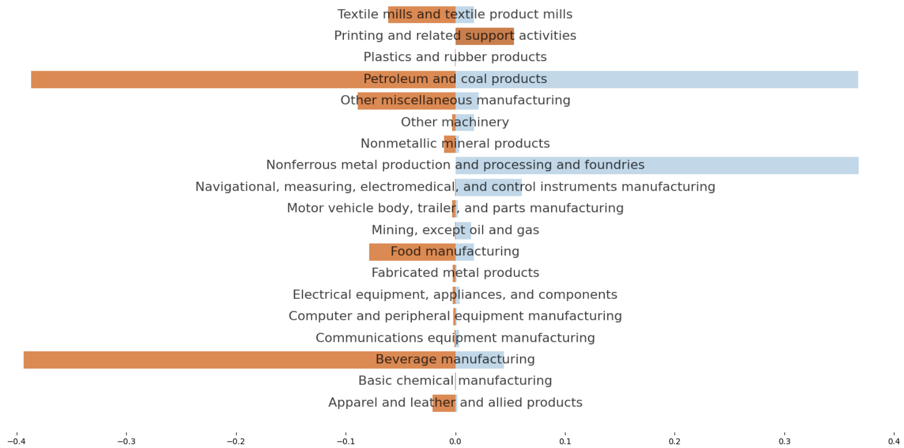


(d) No Oil, No Other

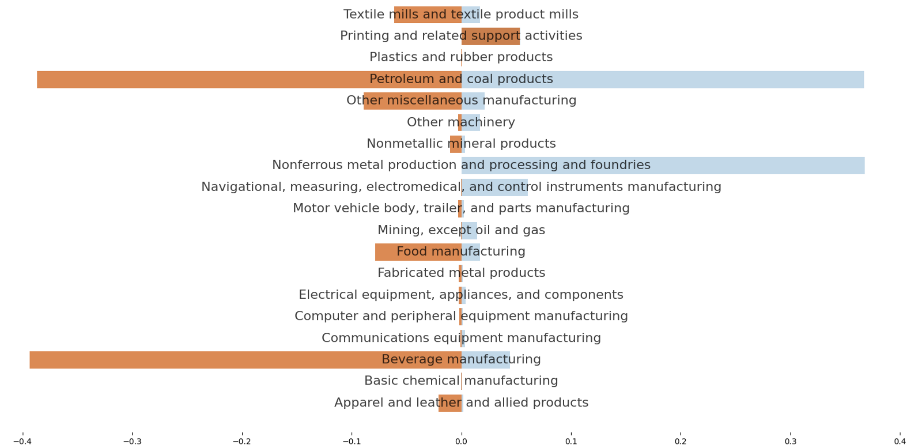
Figure 4: Antigua and Barbuda Imports vs. Direct Effects Shares

(a) Complete

(b) No Oil



(c) No Other



(d) No Oil, No Other

Figure 5: Azerbaijan Imports vs. Direct Effects Shares

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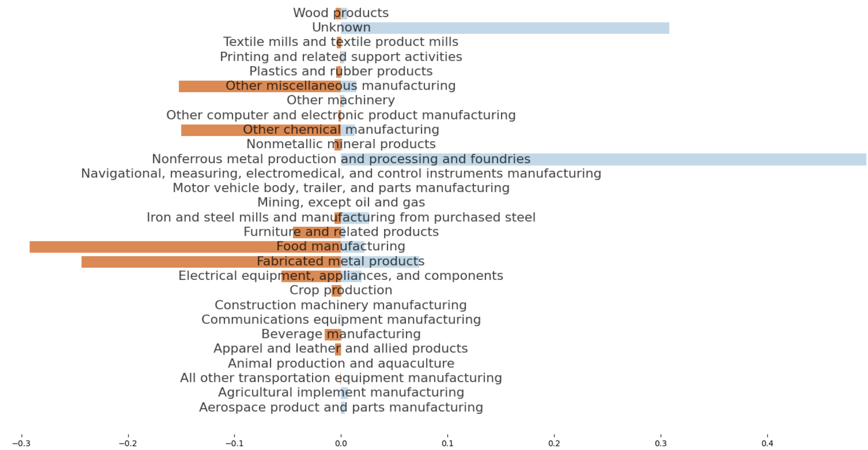
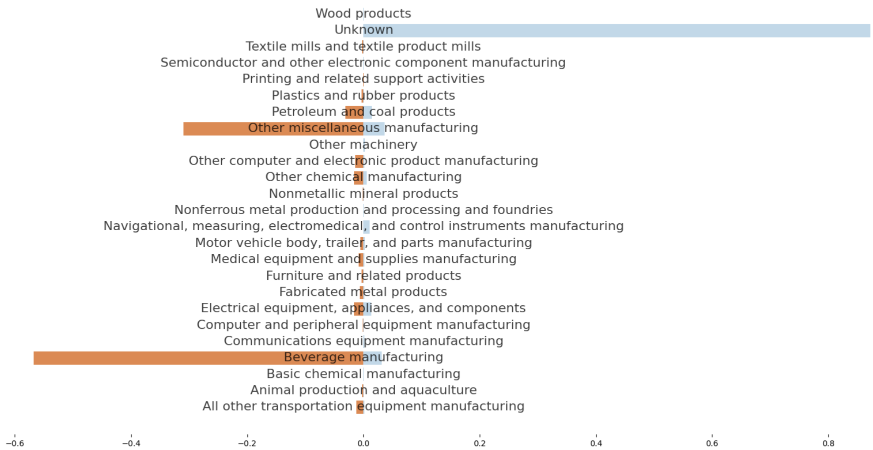


(b) No Oil

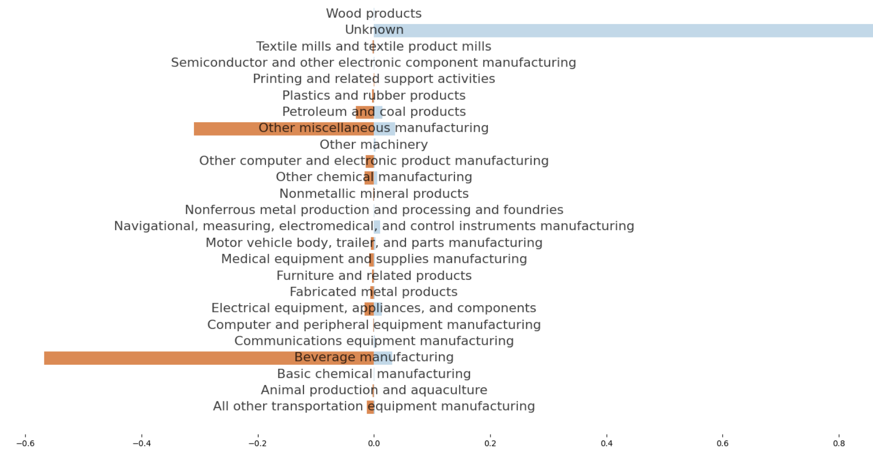
(c) No Other

(d) No Oil, No Other

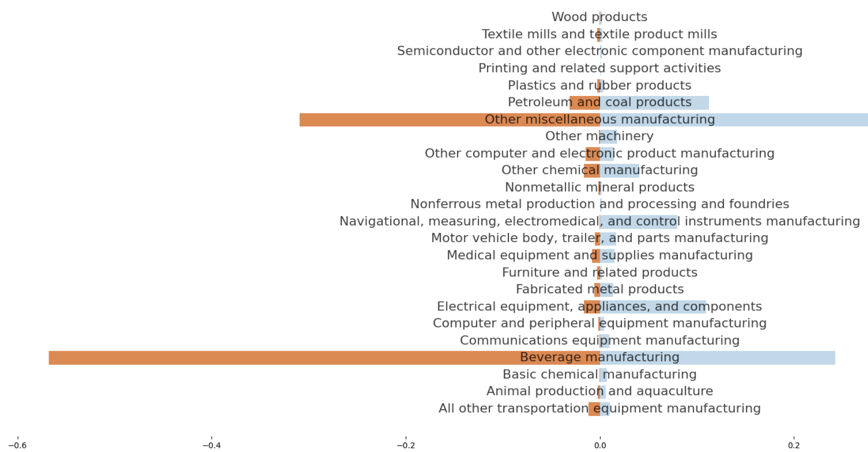
Figure 6: Bermuda Imports vs. Direct Effects Shares



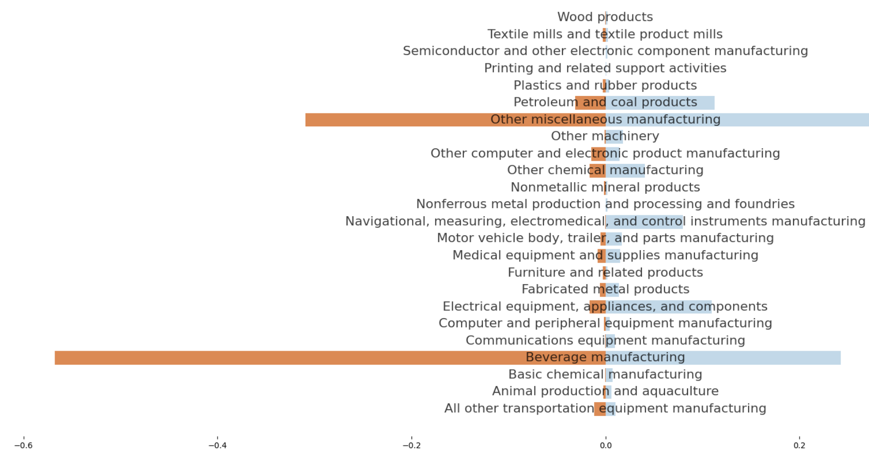
(a) Complete



(b) No Oil

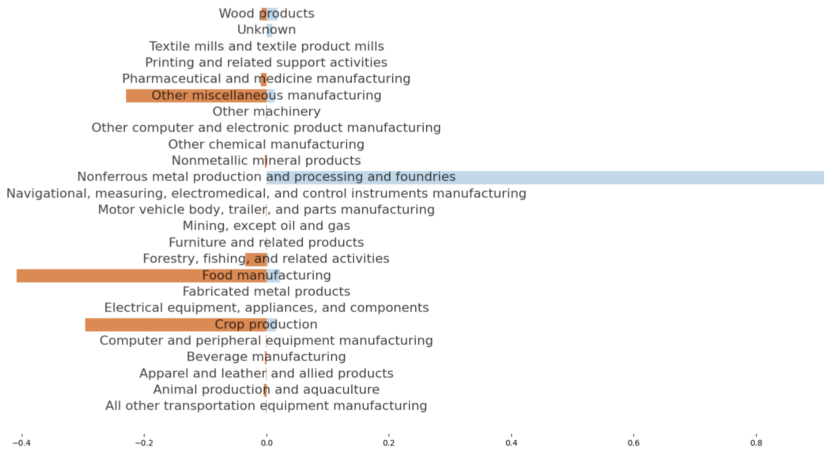


(c) No Other



(d) No Oil, No Other

Figure 7: DR Congo Imports vs. Direct Effects Shares



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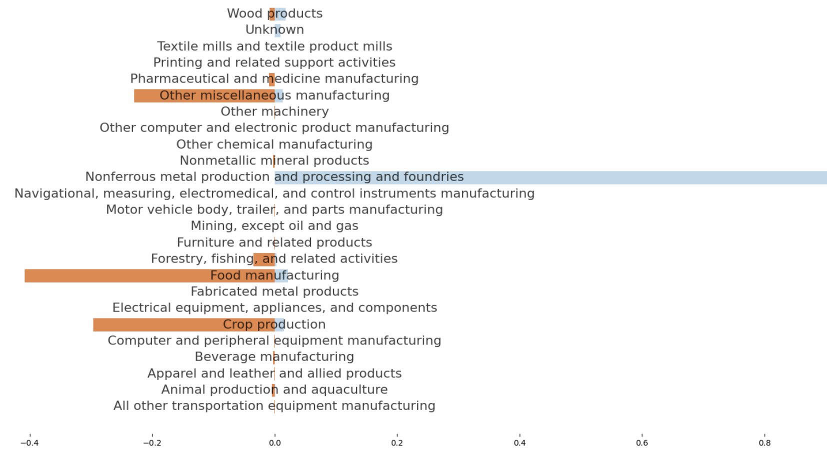
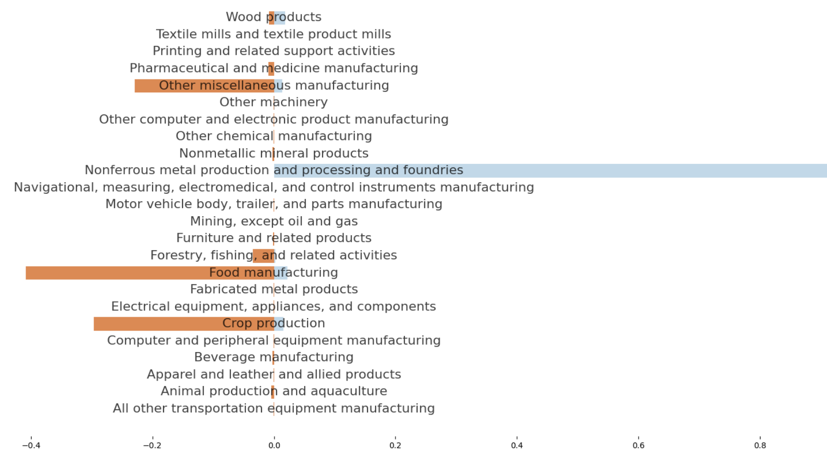


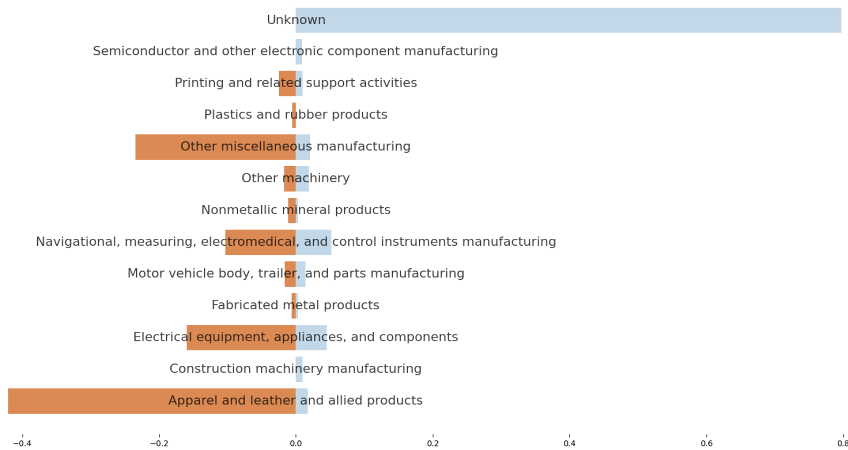
(b) No Oil

(c) No Other

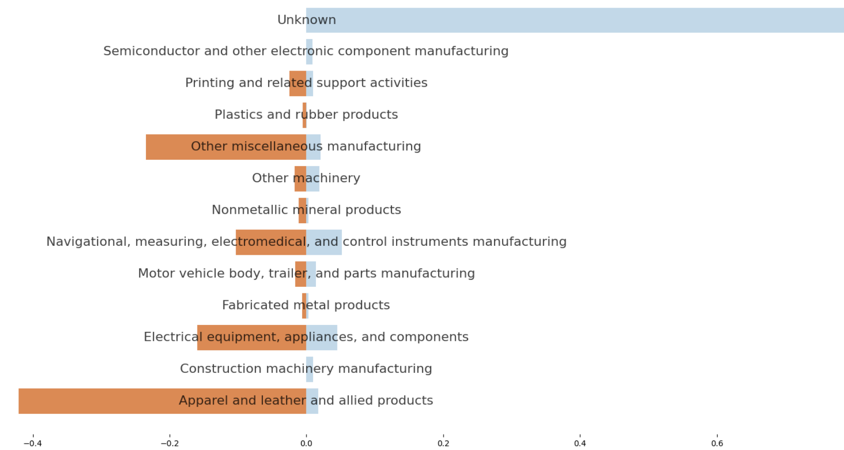


(d) No Oil, No Other

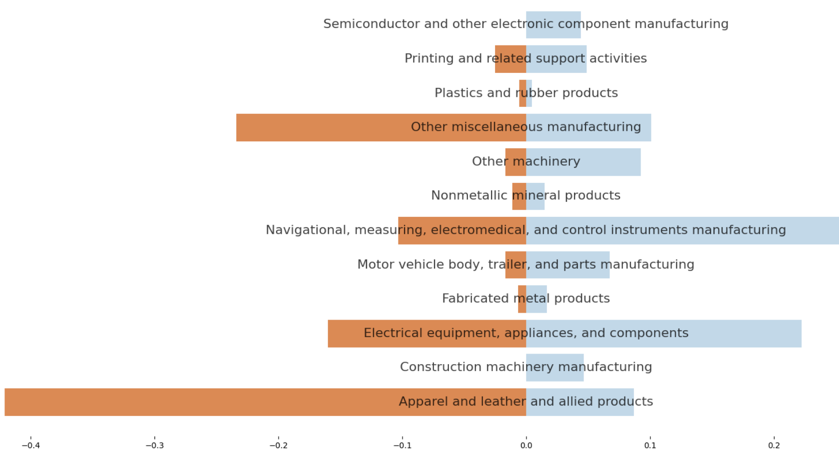
Figure 8: Cook Islands Imports vs. Direct Effects Shares



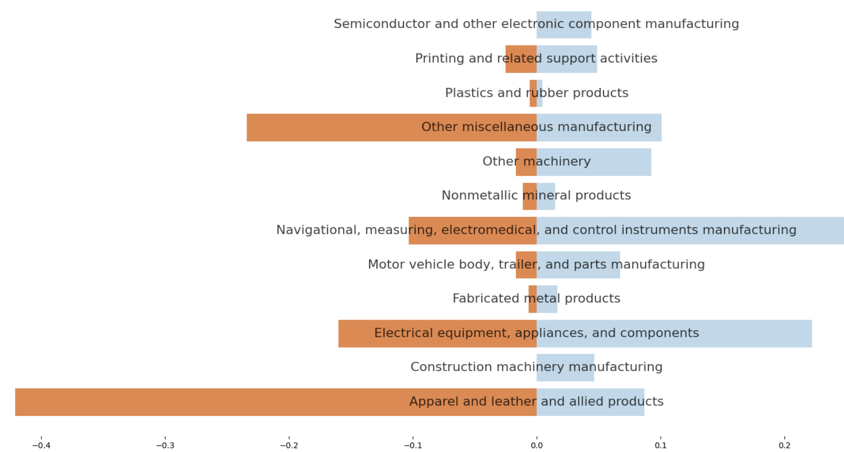
(a) Complete



(b) No Oil

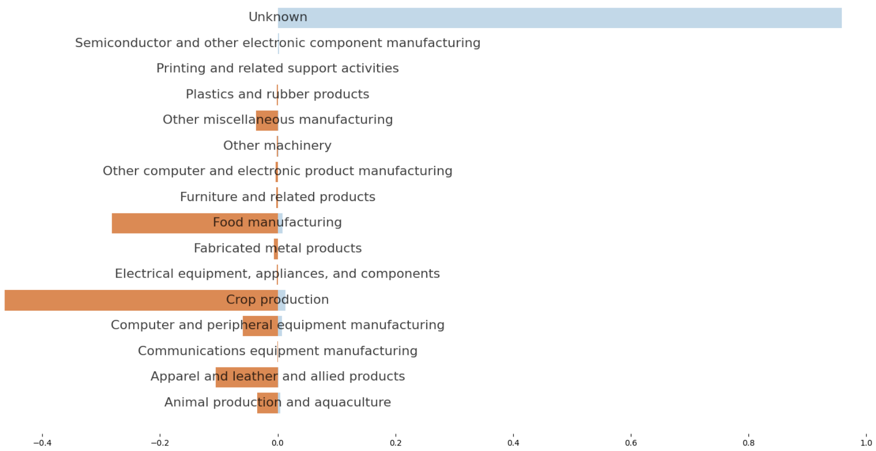


(c) No Other



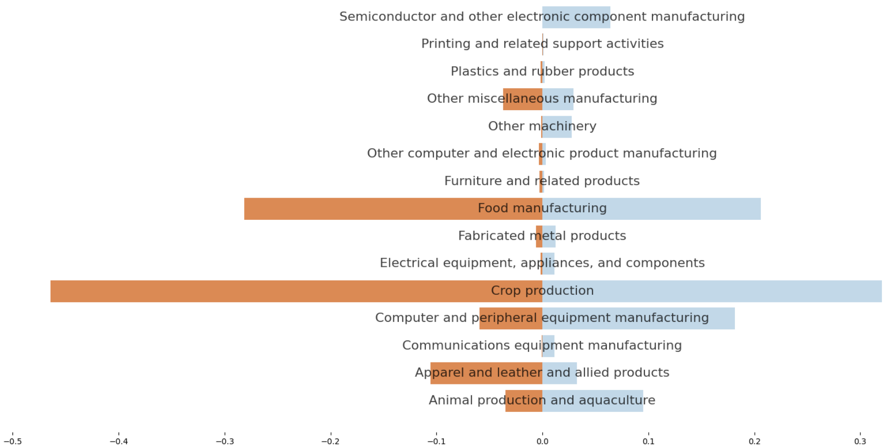
(d) No Oil, No Other

Figure 9: Djibouti Imports vs. Direct Effects Shares



(a) Complete

(b) No Oil



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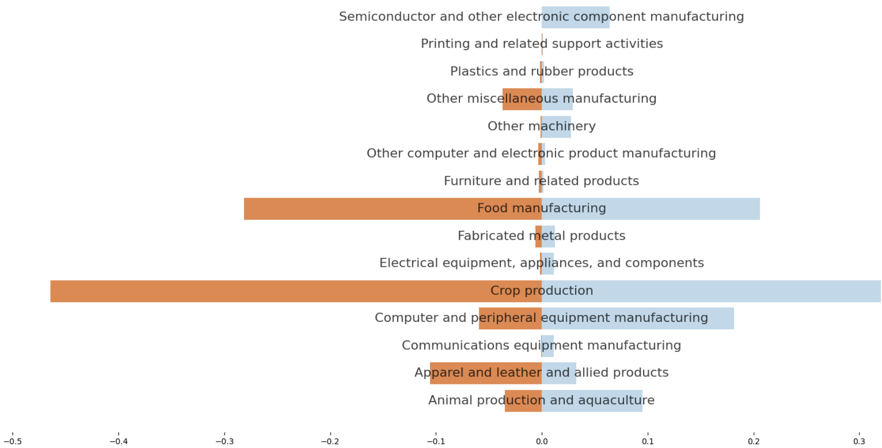
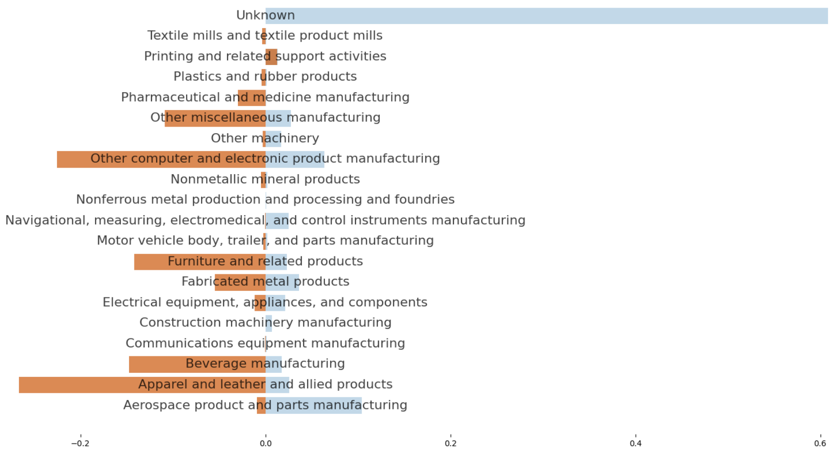
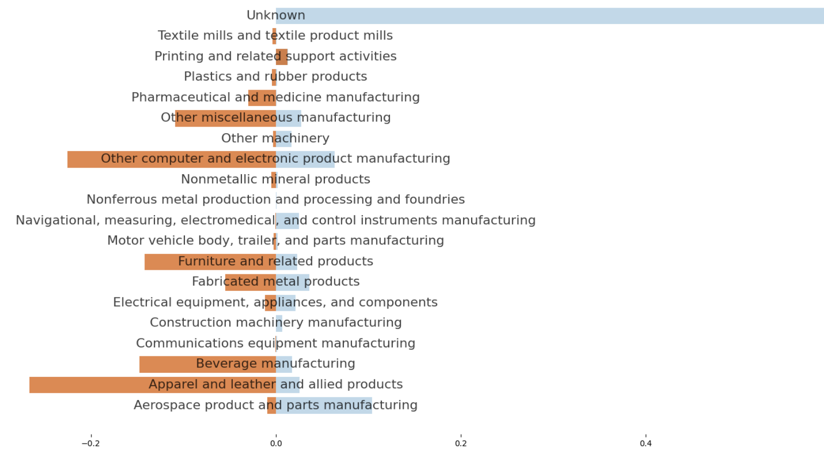


(d) No Oil, No Other

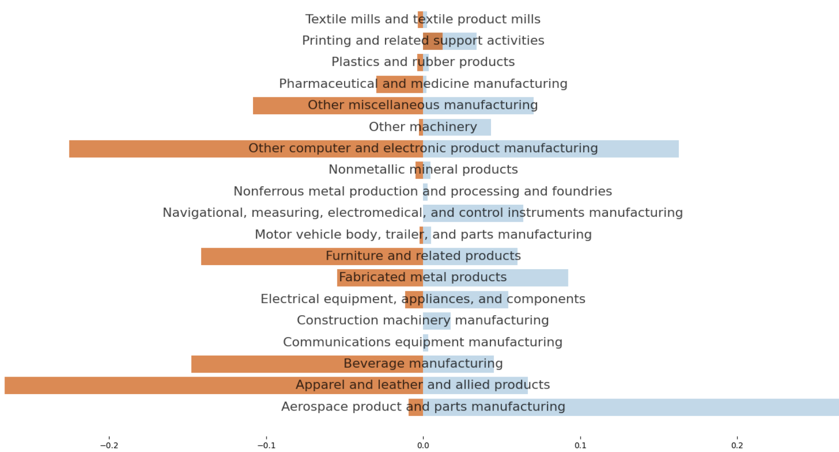
Figure 10: Guadeloupe Imports vs. Direct Effects Shares



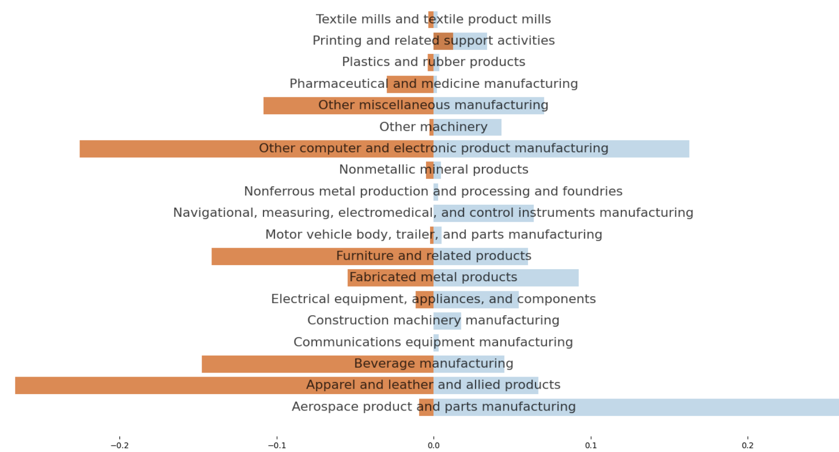
(a) Complete



(b) No Oil

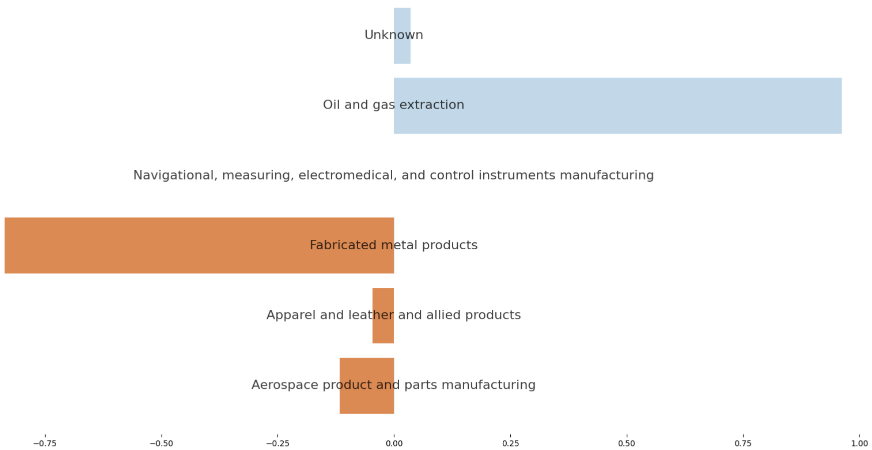


(c) No Other

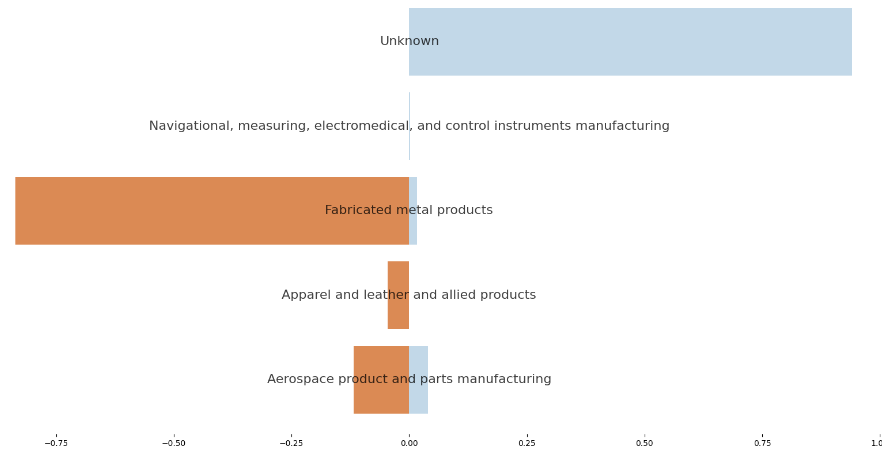


(d) No Oil, No Other

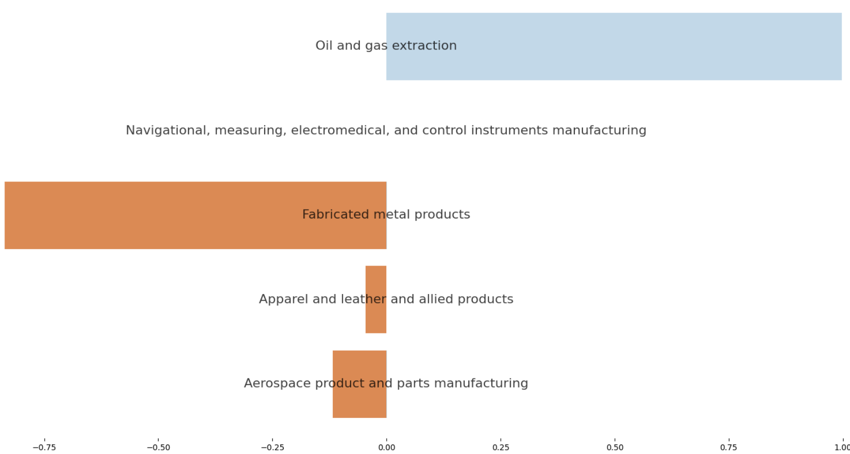
Figure 11: Equatorial Guinea Imports vs. Direct Effects Shares



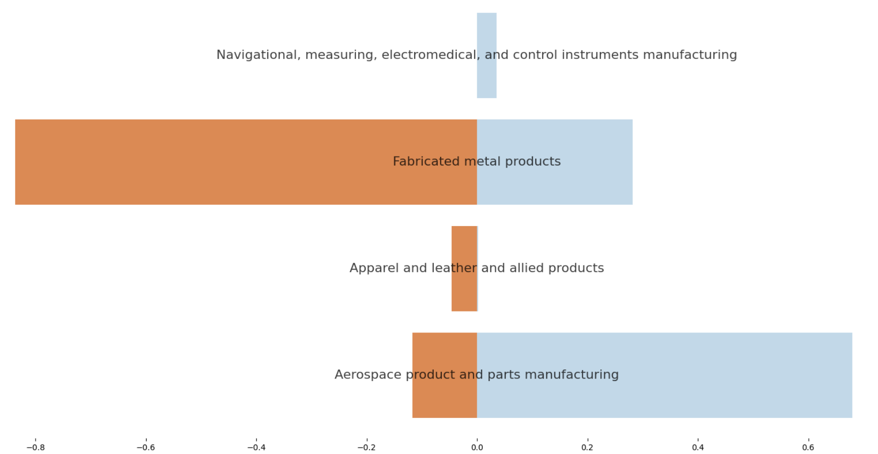
(a) Complete



(b) No Oil



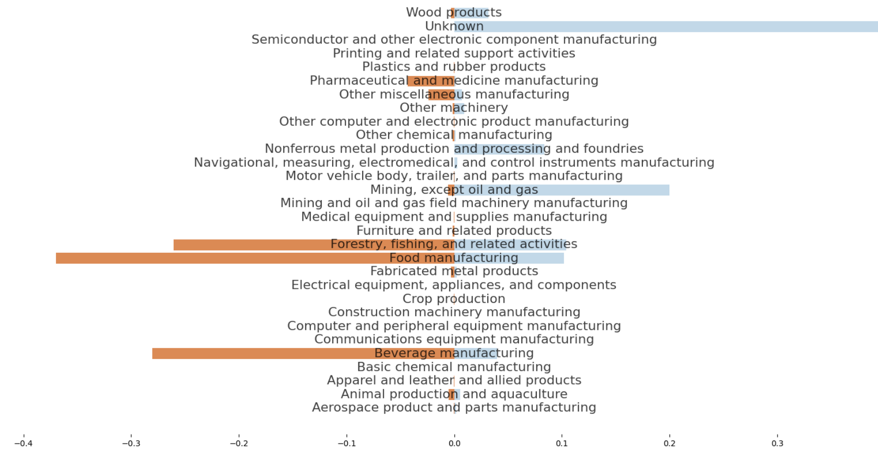
(c) No Other



(d) No Oil, No Other

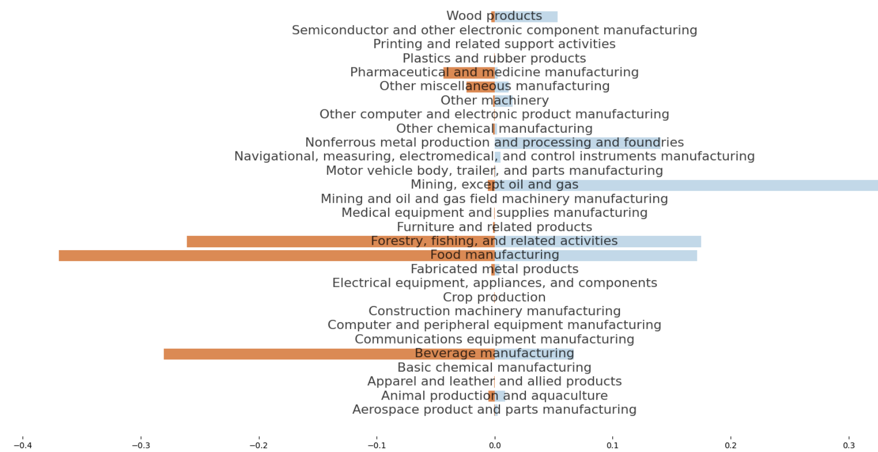
Figure 12: Guyana Imports vs. Direct Effects Shares

(a) Complete



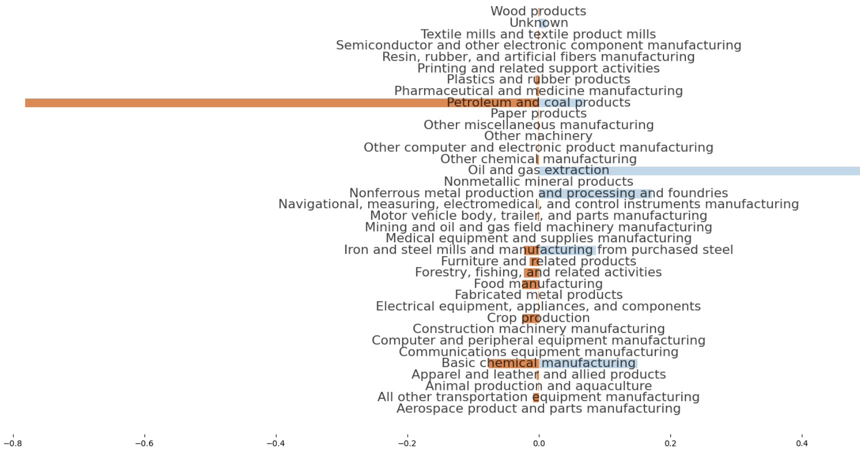
(b) No Oil

(c) No Other



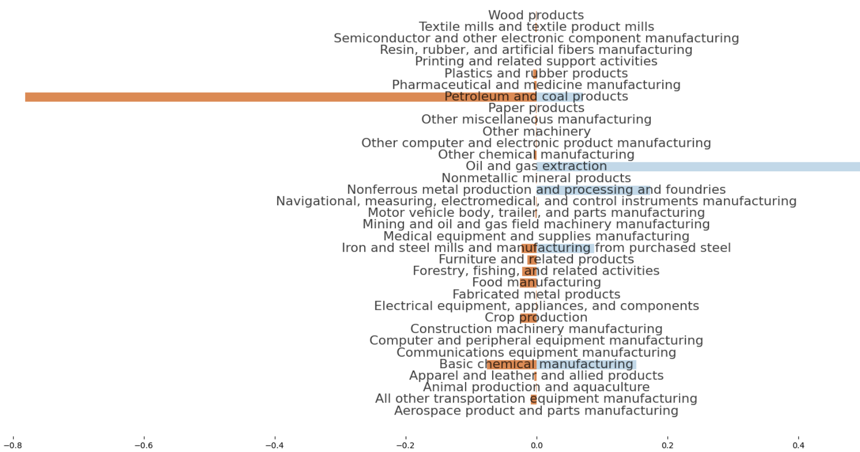
(d) No Oil, No Other

Figure 13: Kazakhstan Imports vs. Direct Effects Shares



(a) Complete

(b) No Oil



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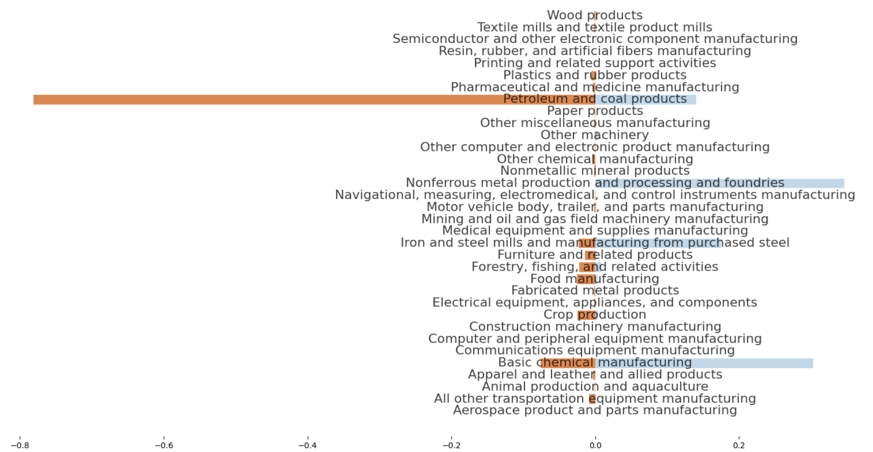
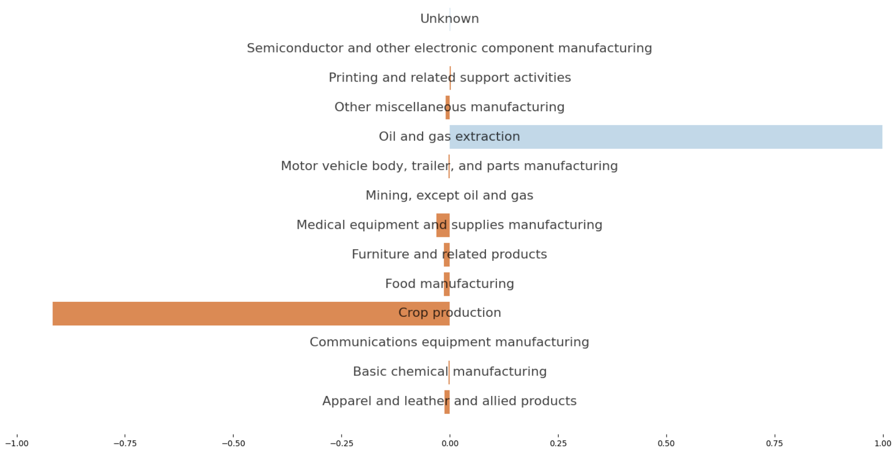
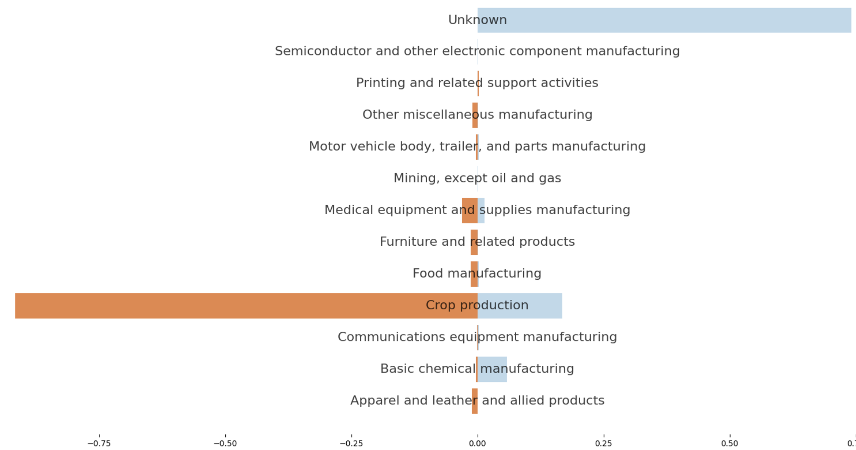


(d) No Oil, No Other

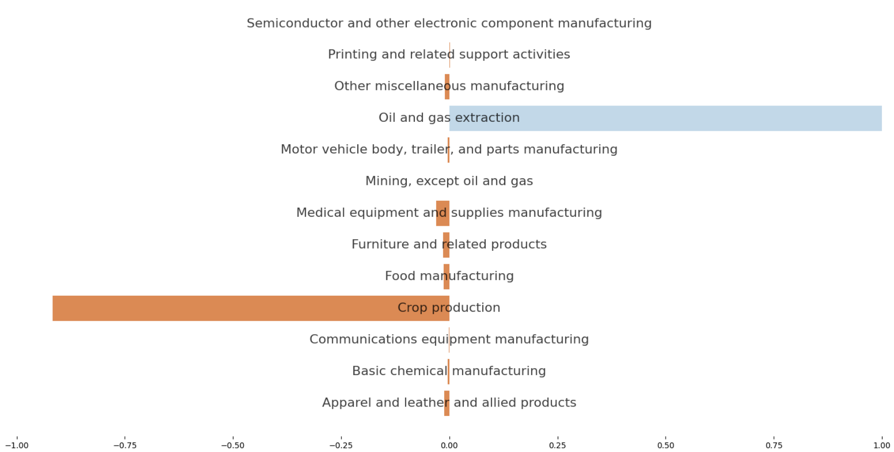
Figure 14: Libya Imports vs. Direct Effects Shares



(a) Complete



(b) No Oil



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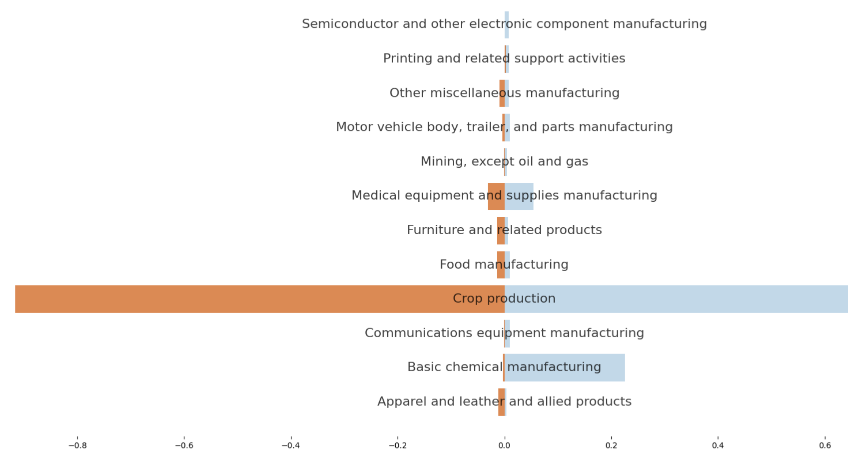
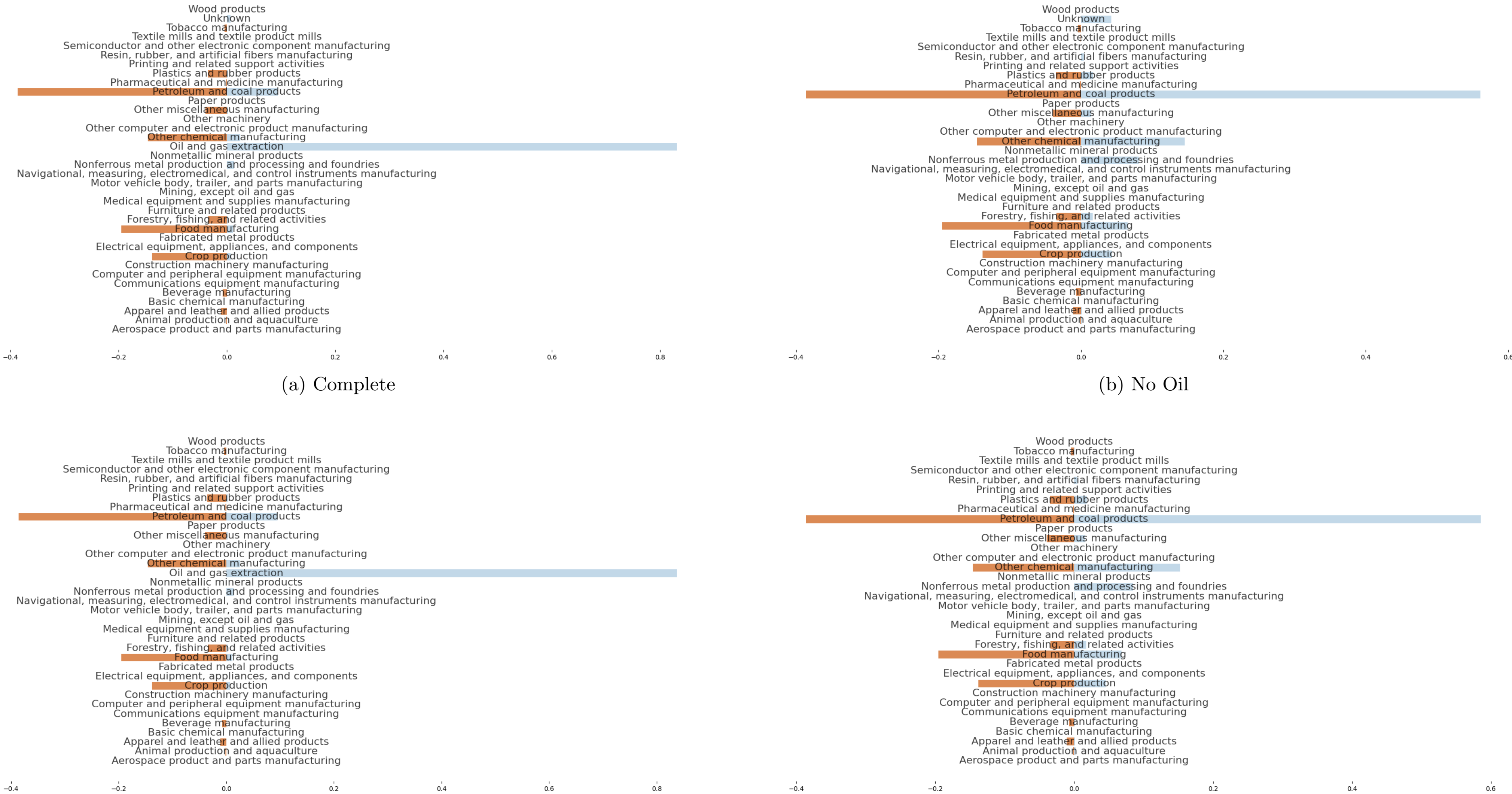


(d) No Oil, No Other

Figure 15: Nigeria Imports vs. Direct Effects Shares



(a) Complete

(b) No Oil

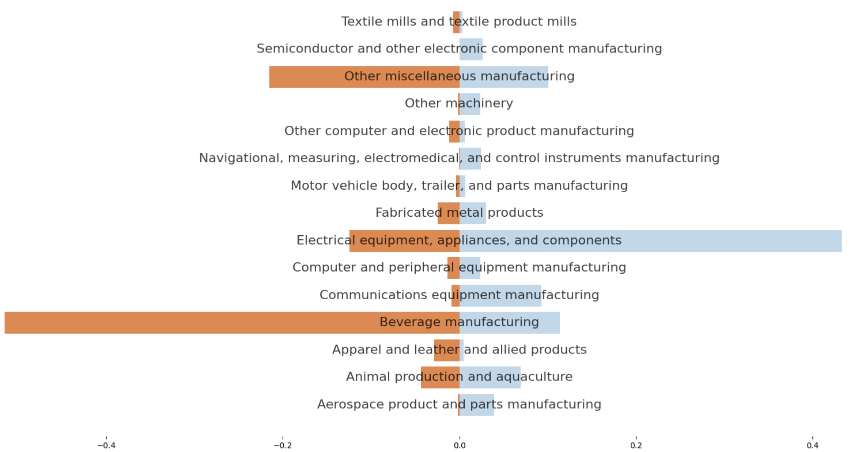
(c) No Other

(d) No Oil, No Other

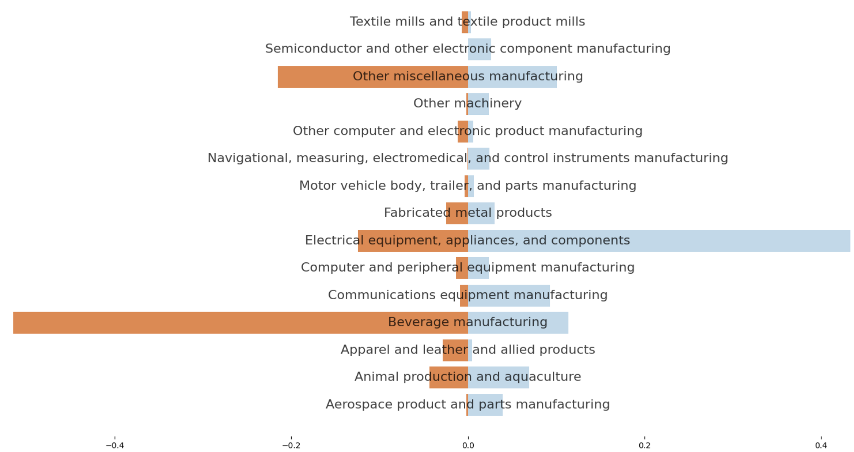
Figure 16: Sint Maarten Imports vs. Direct Effects Shares

(a) Complete

(b) No Oil

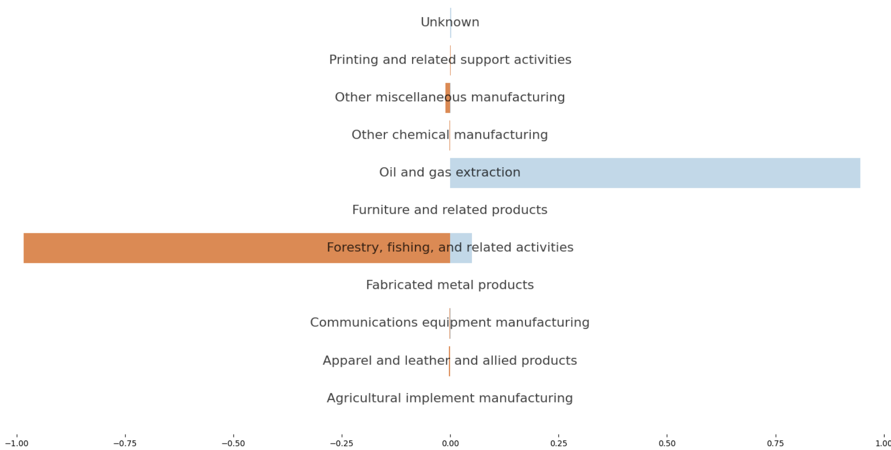


(c) No Other

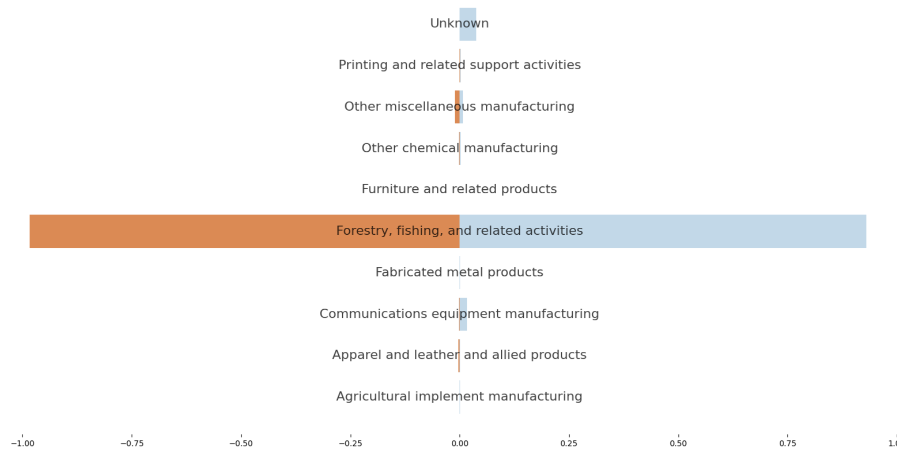


(d) No Oil, No Other

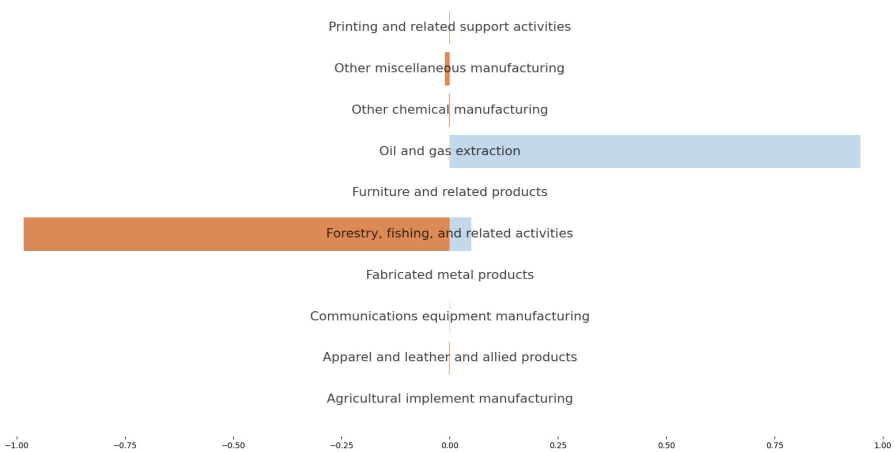
Figure 17: Chad Imports vs. Direct Effects Shares



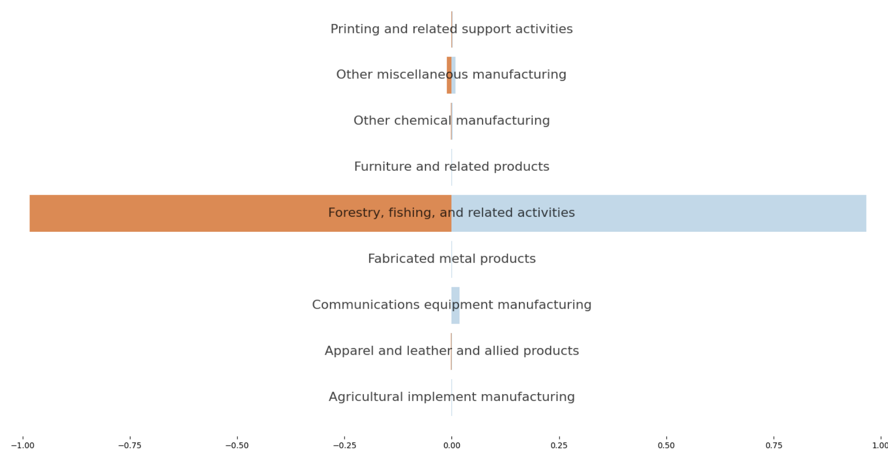
(a) Complete



(b) No Oil



(c) No Other



(d) No Oil, No Other

Figure 18: Tajikistan Imports vs. Direct Effects Shares

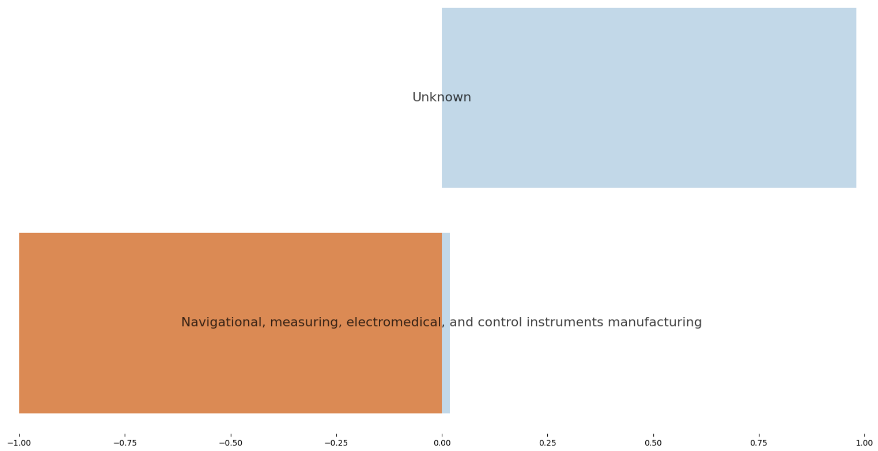
(a) Complete

(b) No Oil

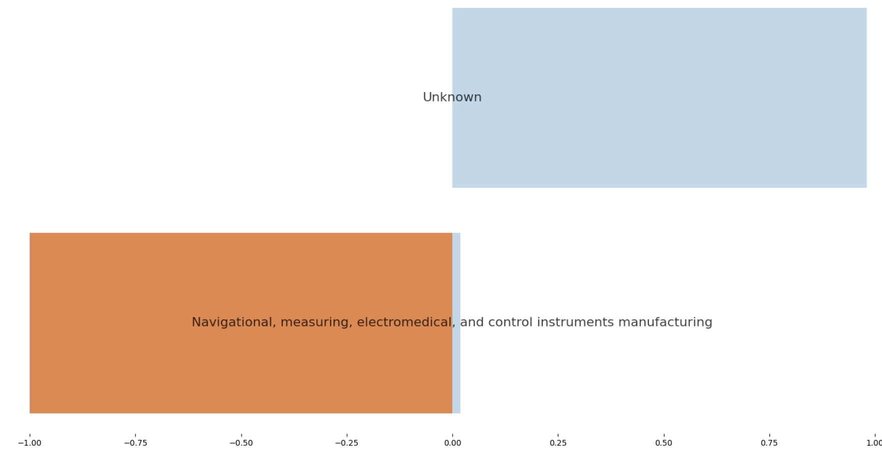
(c) No Other

(d) No Oil, No Other

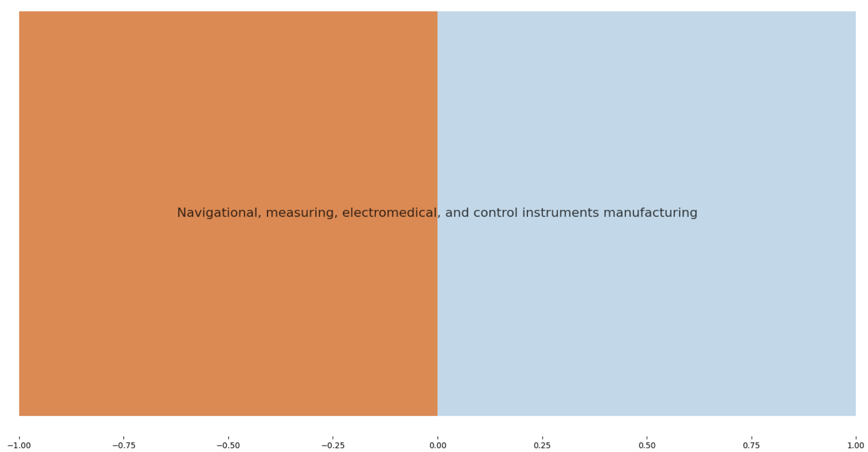
Figure 19: Tokelau Imports vs. Direct Effects Shares



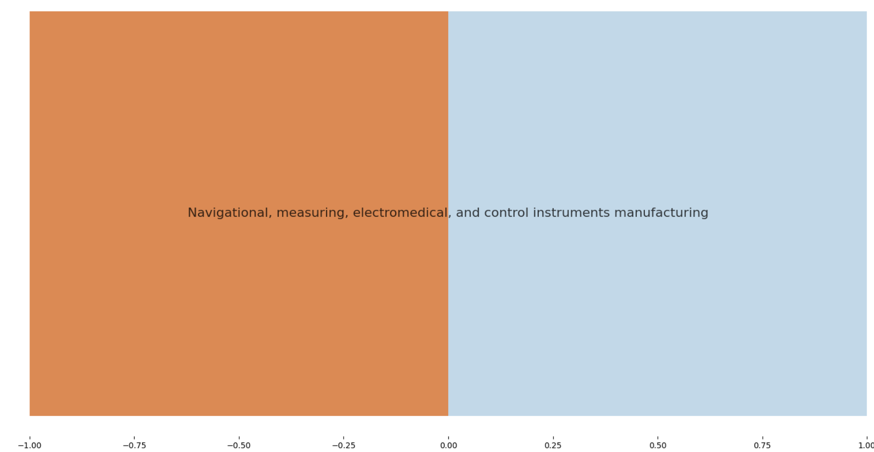
(a) Complete



(b) No Oil



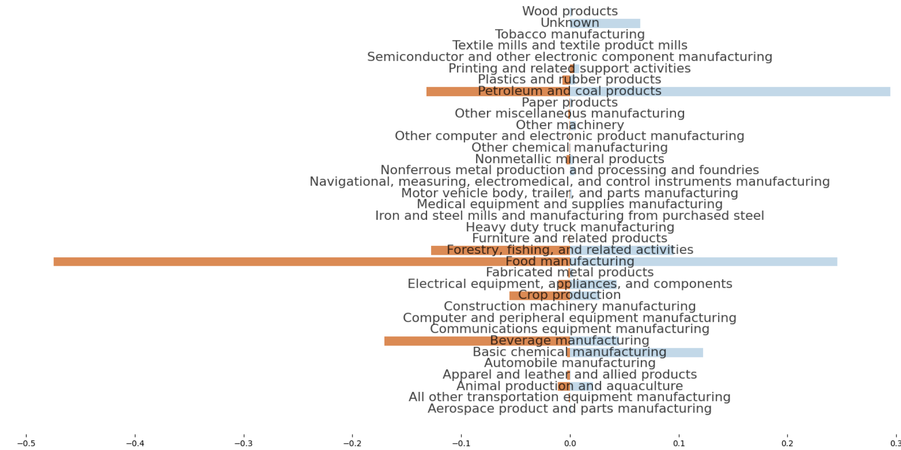
(c) No Other



(d) No Oil, No Other

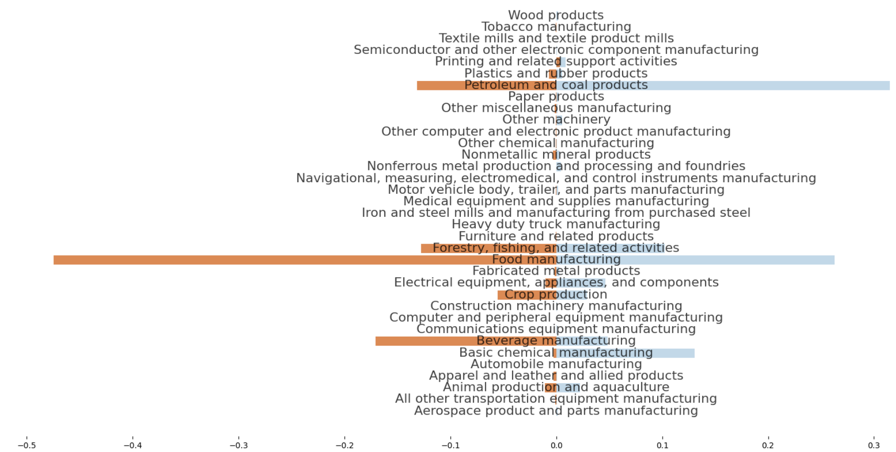
Figure 20: Venezuela Imports vs. Direct Effects Shares

(a) Complete



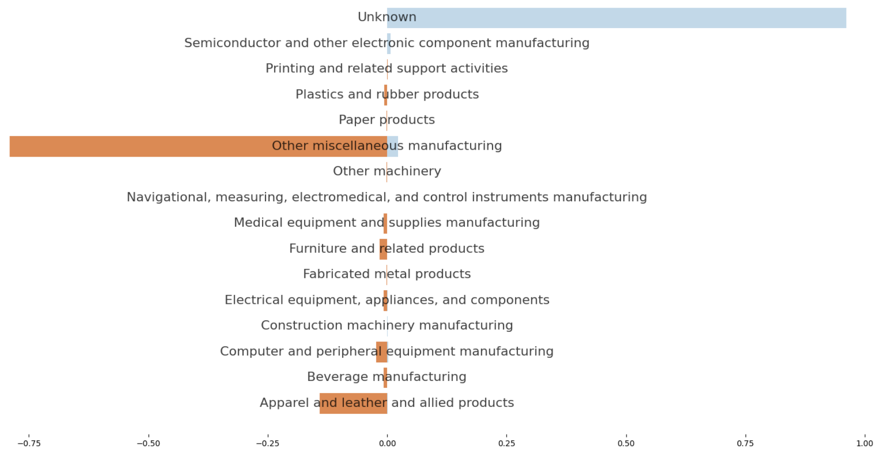
(b) No Oil

(c) No Other

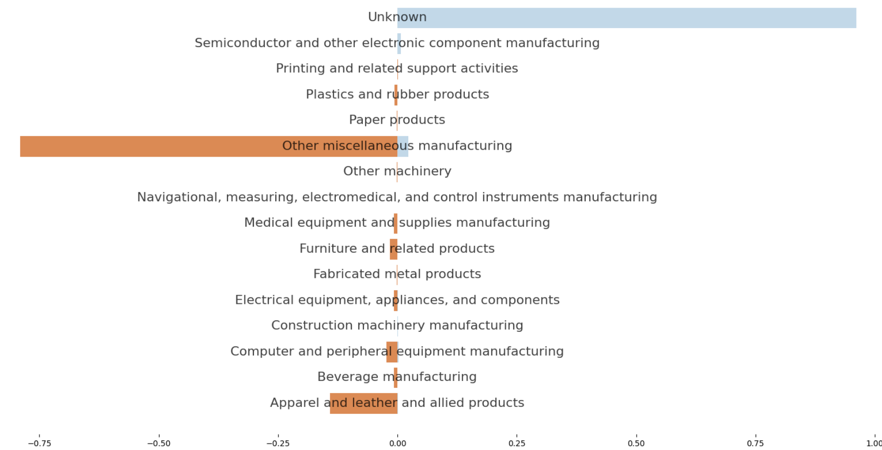


(d) No Oil, No Other

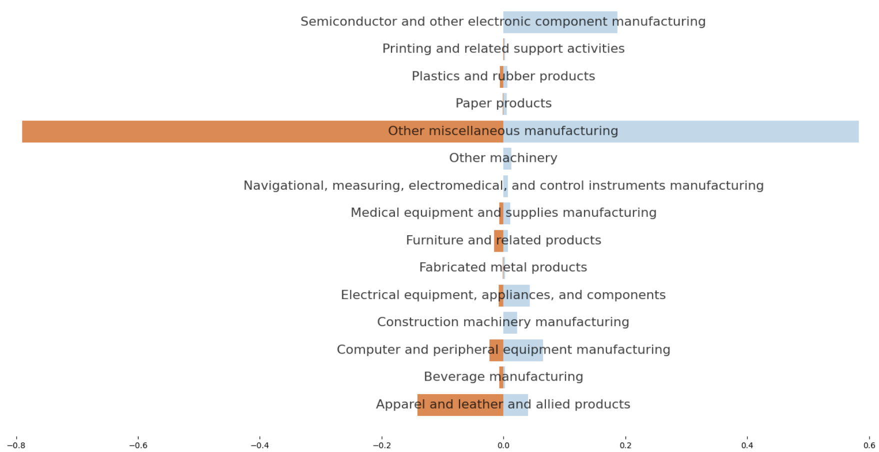
Figure 21: British Virgin Islands Imports vs. Direct Effects Shares



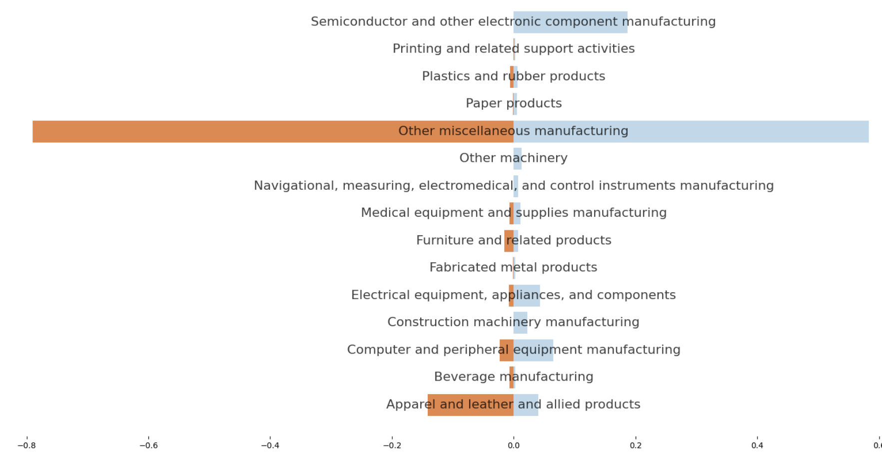
(a) Complete



(b) No Oil



(c) No Other



(d) No Oil, No Other

2 Cluster Concentration Analysis

While cosine similarity provides the preferred analytical approach, this supplementary cluster analysis offers additional perspective on import-effect alignment.

I create two sets of country clusters: one based on import vectors (ordered import values by BEA commodities) and another based on effect vectors (ordered effects by BEA commodities). Effects are calculated under a 10% homogeneous tariff scenario applied to all countries and goods. Figures **??** and **??** display results, with cell numbers representing countries sharing specific import-effect cluster pairs.

Method: I calculate commodity shares of country imports and effects (direct, indirect, total), apply standard scalar normalization (mean 0, unit variance) to prevent large-magnitude variables from dominating clustering, then compute clusters using KMeans. High concentration between import and effect clusters (not necessarily on the main diagonal) indicates alignment between specific import and effect cluster pairs.

Findings: Results demonstrate substantial concentration between import and effect clusters (approximately 75%), as shown in Table **??**.

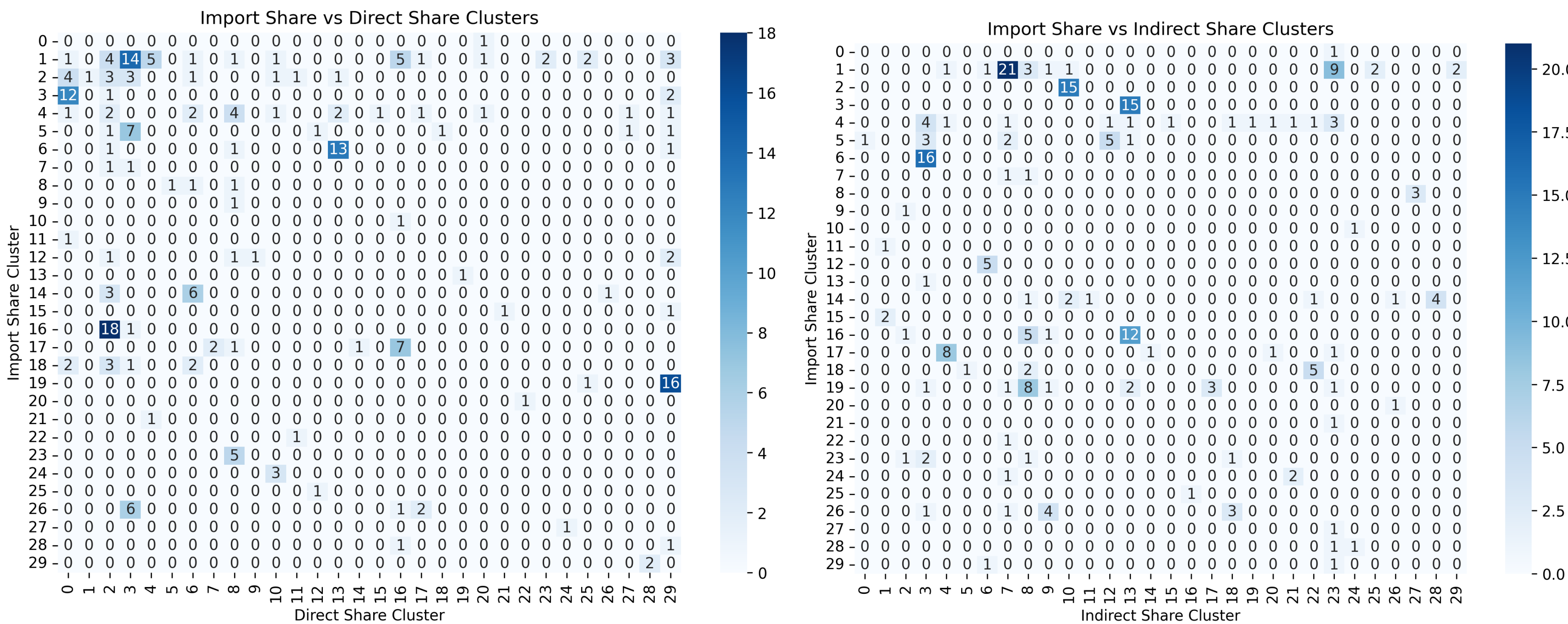


Figure 22: Import vs Direct Effects (left) and Import vs Indirect Effects (right)

3 Case Study: Israel and United Kingdom Analysis

Analysis of ranking comparisons revealed unusual patterns for the UK and Israel. The UK exports predominantly high consumer-value products (beverages, automobiles, foods, pharmaceuticals), while Israel, despite exporting some high consumer-value goods (pharmaceuticals, miscellaneous manufacturing), also exports substantial quantities of non-consumer goods (communications, semiconductor manufacturing). This imbalance between the countries reflects differences in PCE composition of goods dominating their respective import structures to the US.

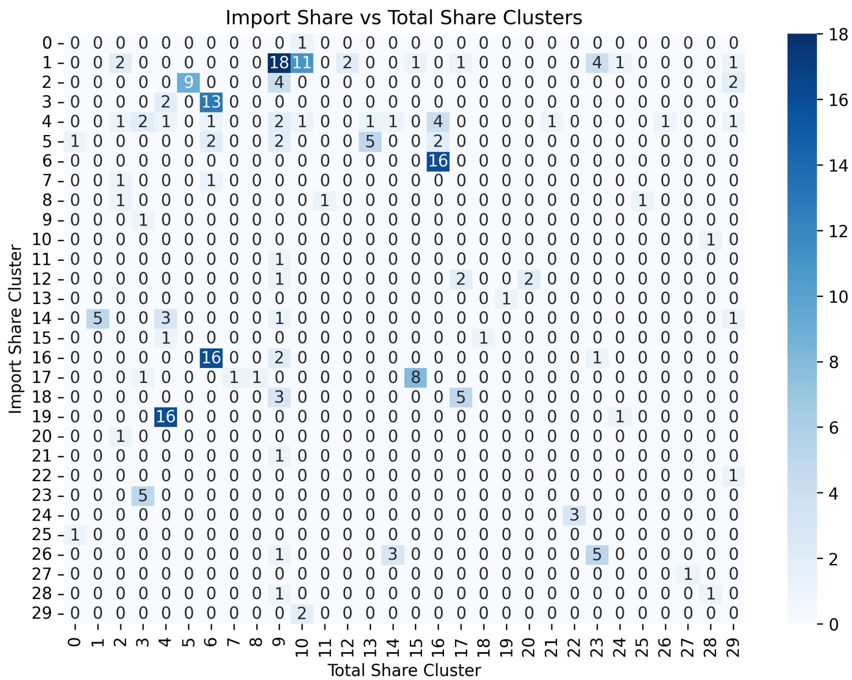
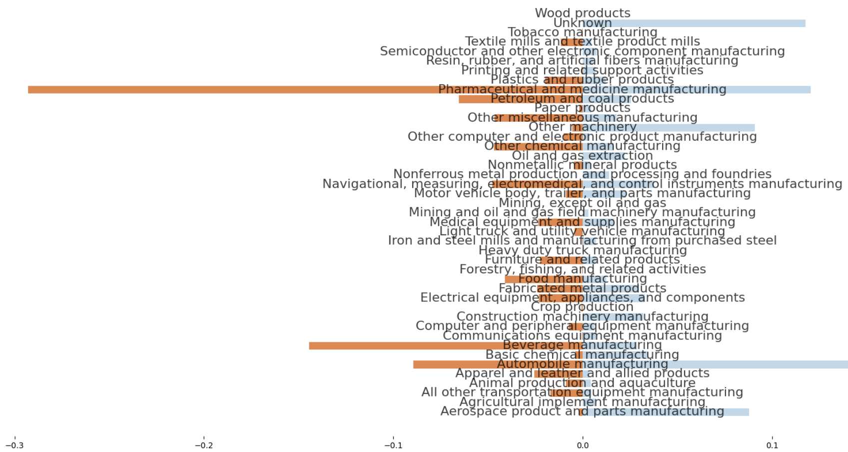


Figure 23: Import vs Total Effects

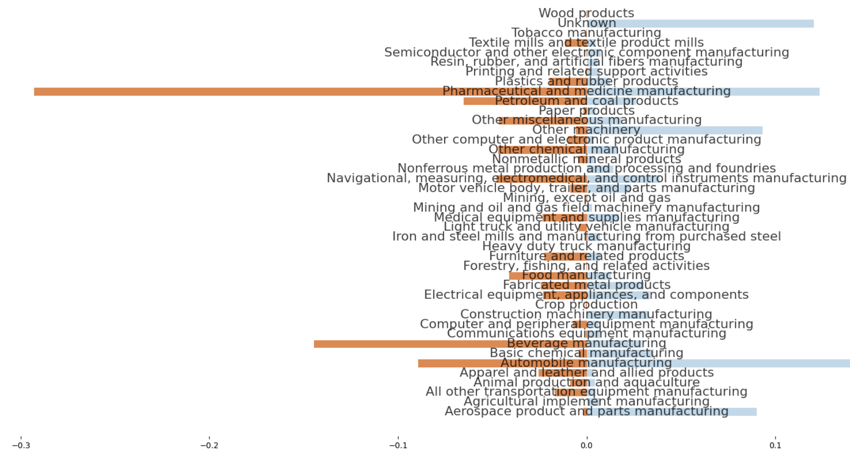
Table 3: Cluster Concentration Analysis - 10% Tariff Scenario

Cluster	Direct		Indirect		Total	
	Import→Effect	Effect→Import	Import→Effect	Effect→Import	Import→Effect	Effect→Import
0	100.0%	57.1%	100.0%	100.0%	100.0%	50.0%
1	34.1%	100.0%	51.2%	66.7%	43.9%	100.0%
2	26.7%	47.4%	100.0%	33.3%	60.0%	33.3%
3	80.0%	42.4%	100.0%	57.1%	86.7%	55.6%
4	23.5%	83.3%	23.5%	80.0%	23.5%	69.6%
5	58.3%	100.0%	41.7%	100.0%	41.7%	100.0%
6	81.2%	46.2%	100.0%	71.4%	100.0%	48.5%
7	50.0%	100.0%	50.0%	72.4%	50.0%	100.0%
8	33.3%	33.3%	100.0%	38.1%	33.3%	100.0%
9	100.0%	100.0%	100.0%	57.1%	100.0%	48.6%
10	100.0%	50.0%	100.0%	83.3%	100.0%	73.3%
11	100.0%	50.0%	100.0%	100.0%	100.0%	100.0%
12	40.0%	50.0%	100.0%	83.3%	40.0%	100.0%
13	100.0%	81.2%	100.0%	48.4%	100.0%	83.3%
14	60.0%	100.0%	40.0%	100.0%	50.0%	75.0%
15	50.0%	100.0%	100.0%	100.0%	50.0%	88.9%
16	94.7%	46.7%	63.2%	100.0%	84.2%	72.7%
17	63.6%	50.0%	72.7%	100.0%	72.7%	62.5%
18	37.5%	100.0%	62.5%	60.0%	62.5%	100.0%
19	94.1%	100.0%	47.1%	100.0%	94.1%	100.0%
20	100.0%	33.3%	100.0%	50.0%	100.0%	100.0%
21	100.0%	100.0%	100.0%	66.7%	100.0%	100.0%
22	100.0%	100.0%	100.0%	71.4%	100.0%	100.0%
23	100.0%	100.0%	40.0%	47.4%	100.0%	50.0%
24	100.0%	100.0%	66.7%	50.0%	100.0%	50.0%
25	100.0%	66.7%	100.0%	100.0%	100.0%	100.0%
26	66.7%	100.0%	44.4%	50.0%	55.6%	100.0%
27	100.0%	50.0%	100.0%	100.0%	100.0%	100.0%
28	50.0%	100.0%	50.0%	100.0%	50.0%	50.0%
29	100.0%	57.1%	50.0%	100.0%	100.0%	33.3%
Average	74.8%	74.8%	76.8%	76.2%	76.6%	78.2%

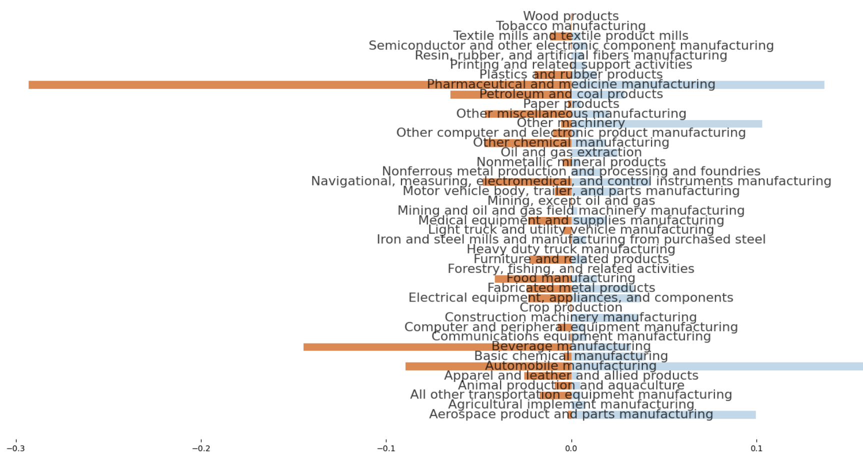
Figure 24: United Kingdom Imports vs. Direct Effects Shares



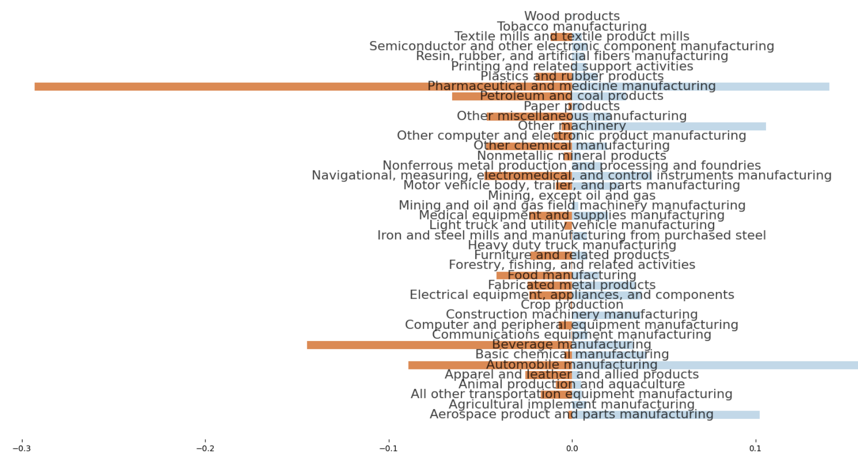
(a) Complete



(b) No Oil

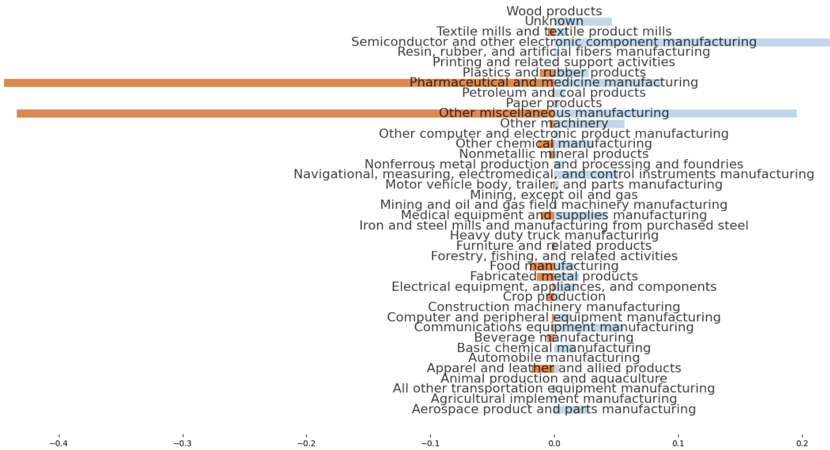


(c) No Other

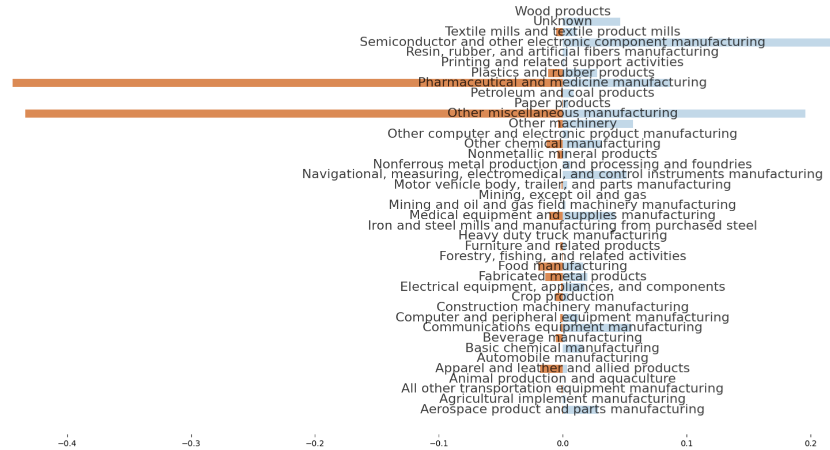


(d) No Oil, No Other

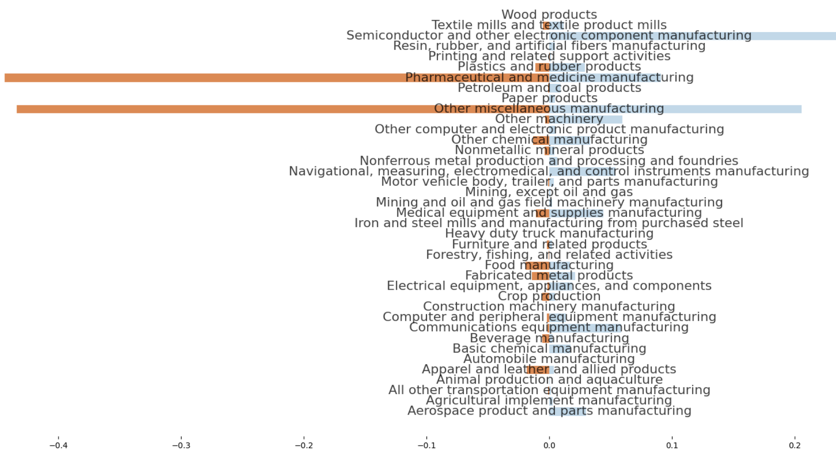
Figure 25: Israel Imports vs. Direct Effects Shares



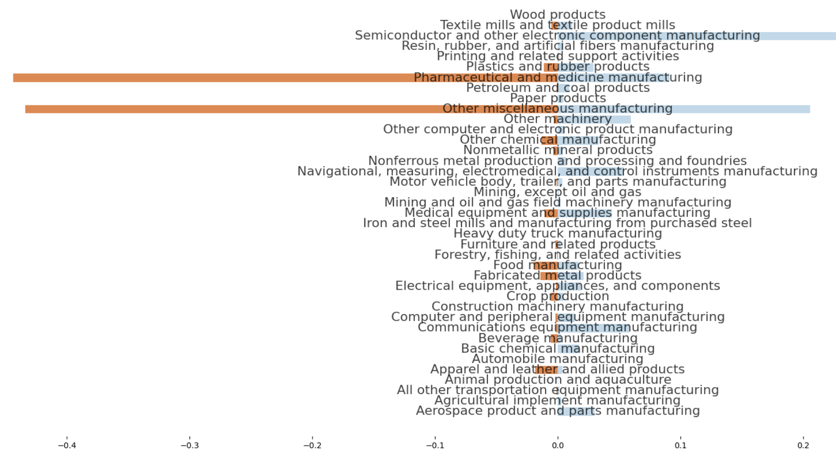
(a) Complete



(b) No Oil



(c) No Other



(d) No Oil, No Other